

HUY LAI

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TECHNICAL SKILLS

Languages: Java, Python, C/C++, Ruby, SQL, JavaScript, TypeScript, HTML, CSS

Frameworks & Libraries: Spring Boot, React.js, Next.js, Node.js, Ruby on Rails, JUnit, Mockito, PyTorch, TensorFlow, Scikit-Learn

Data & Messaging: Apache Kafka, OpenSearch, Elasticsearch, PostgreSQL, Oracle, H2, AWS S3, Jackson

DevOps & Tools: Git, Docker, Maven, AWS, GCP, Nginx, Prometheus, Kibana, Firebase, Pub/Sub, VS Code, IntelliJ

EXPERIENCE

Software Engineer | [Visa Inc.](#) ; *Austin, TX* | *Aug 2025 — Present*

- Built an indexing library in Java with resilient OpenSearch/In-Memory storage and byte-level S3 range optimization for sub-100ms audit log lookups.
- Engineered a Kafka MCP Server to automate resource provisioning via Jira, GitHub, and Wiki integration, reducing developer onboarding time from weeks to days.
- Built a LangGraph agent that automates runtime configuration toggles provisioning from Jira tickets across 6 environments, replacing a 10-step manual workflow with real-time metadata validation and structured Jira feedback.
- Diagnosed and escalated critical bugs, enhancing system reliability and browser compatibility.

Software Engineer | [Total Care 2U](#) ; *Waxahachie, TX* | *Jul 2024 — Jul 2025*

- Automated the end-to-end ETL (Extract, Transform, Load) process for migrating patient medical data into an advanced Electronic Health Record (EHR) system.
- Validated patients' personal information for accuracy and completeness to ensure data integrity.
- Designed automated quality assurance procedures to verify data consistency across systems.
- Presented progress demos to cross-functional teams to ensure stakeholder alignment and gather feedback.

Product Owner and Developer | [BVCOG](#) ; *Bryan, TX* | *Aug 2023 — Dec 2023*

- Spearheaded the development of a scalable full-stack application using Ruby on Rails, capable of managing over 400 contracts simultaneously.
- Implemented and rigorously tested 85+ individual features using Cucumber and RSpec to ensure high-quality code delivery.
- Facilitated requirement gathering as primary liaison between developers and stakeholders, driving iterative feedback loops.

EDUCATION

Texas A&M University | College Station, TX | May 2025 | GPA: 3.700 / 4.000

Master of Computer Science

Texas A&M University | College Station, TX | May 2024 | GPA: 3.594 / 4.000

Bachelor of Science in Computer Science, cum laude, Minors: Cybersecurity, Mathematics

PROJECTS

Improving CLIP Training | *torch, Self-Supervised Learning, Contrastive Learning* | Aug 2024 — Dec 2024

- Optimized bimodal contrastive SSL models using global contrastive loss for scalable image-text learning.
- Implemented SogCLR/iSogCLR, enhancing retrieval accuracy and zero-shot classification.
- Analyzed optimizers and loss functions to improve model convergence and multimodal generalization.

ScriptSearch | *Python, Cloud Functions, Firebase, Pub/Sub, Typesense* | Jan 2024 — May 2024

- Built a transcript-based YouTube search engine with fast, keyword-driven results and timestamped links.
- Engineered a scalable scraping pipeline using Golang, Pub/Sub, and Firebase Functions.
- Achieved sub-4s average query time across 250+ videos using Typesense indexing.
- Delivered a full-stack app in 15 weeks through Agile development and user-driven design.